Joined lane-drop VSL candidate: contract and qualification

2026-10-21 — LF1/LF2 selected; concrete member delivered at JQ1

Reading map. Pages 1–2 specify the traffic member; pages 3–6 give information, complete-accounting and derivative qualifications. These are analytical results and declared defaults, not simulator or traffic experiments. The earlier accepted VSL report is unchanged.

1. Concrete member and global assumptions

There are $N = 10$ road rows of length $\Delta x = 0.1$ km, each with lanes E, A, B in that left-to-right order. At the downstream boundary E ends; A and B have exit outlets. Traffic completes when it crosses that line. There are no physical downstream cells or recovery zone in this member. The whole finite cohort is present at time zero; later external inflow is zero. Its prescribed initial continuous stocks do not depend on the advice. Lane stocks and incentives retain the principal framework [1], with the explicit budget/allocation substitutions below.

All lanes use nominal free speed $u_0 = 60$ km/h, backward wave-speed magnitude $w = 20$ km/h, jam density $k_J = 150$ veh/km and maximum capacity-drop fraction $\delta = 0.1$. Let $\Delta t = 1/600$ h (6 s). One common constant advice $v \in [30, 60]$ km/h applies throughout the modeled road. These are illustrative review defaults, not field calibration or an experimentally selected regime. Use continuous vehicle-equivalent stock $n_{i\ell}^k = \Delta x\, \rho_{i\ell}^k$, $0 \le n_{i\ell}^0 \le J = \Delta x k_J$; time index is k.

2. Selected H budget and fixed boundaries

Define the common FD quantities and Han recovery slope [2, 3]:

$$\begin{aligned} C &= \frac{u_0 w k_J}{u_0 + w} = 2250, & \rho_c &= C/u_0 = 37.5, \\ Q_J &= (1-\delta)C, & w_d &= \frac{Q_J}{k_J - Q_J/u_0} = \frac{540}{31}. \end{aligned} \tag{1}$$

In rates (veh/h), use

$$\begin{aligned} Q(x) &= \min\left\{C, C\left(1 - \delta\frac{x-\rho_c}{k_J-\rho_c}\right)\right\}, \\ R(x,y) &= \min\{Q(x),\ w(k_J - y),\ w(k_J - x) + w_d(x-y)\}, \\ D_{i\ell} &= \min\{v\rho_{i\ell}, Q(\rho_{i-1,\ell})\}. \end{aligned} \tag{2}$$

Set $\rho_{0\ell} = 0$ for the no-inflow upstream boundary. For a target lane j in row $i+1 \le N$, its *one shared* receiving amount is $r_{ij} = \Delta t R(\rho_{ij}, \rho_{i+1,j})$. At $i = N$, only $j = A, B$ exist; set $r_{Nj} = \Delta t R(\rho_{Nj}, 0)$ using free outlet ghost densities. Ghosts are boundary values, not stocks or additional traffic. Advice alters sending, not Q, R, priority weights, geometry or nominal ρ_c directly.

All movements advance one row, diagonally or in the same lane. Evaluate every flow from the old time level and then update all physical stocks together. At interior boundaries allow every adjacent movement and the same-lane movement. At N, E can only enter A; A/B may enter their own or the other continuing outlet. This is the selected flow-realization adaptation, not literal reuse of the principal paper's adjusted-flow formulas.

The selected H proof gives $0 \le D_{i\ell} \le v\rho_{i\ell}$ and $0 \le R(x,y) \le w(k_J - y)$. Because $u_0\Delta t \le \Delta x$ and $w\Delta t \le \Delta x$, demand amounts $d_{i\ell} = \Delta t D_{i\ell}$ cannot exceed stock and receiving amounts cannot exceed old available storage. The allocator below therefore preserves nonnegative stocks, storage, permissions and mass. Native calibration and wave fidelity do not follow from these inequalities.

3. Explicit lane preferences and their local conventions

Use the principal source's density/incentive framework [1] and same-author lookahead weights $(a_0, a_1, a_2) = (2, 2, 1)$ [4]:

$$K_{i\ell} = \frac{\sum_{r\in\mathcal{H}_{i\ell}} a_r \rho_{i+r,\ell}}{\sum_{r\in\mathcal{H}_{i\ell}} a_r}, \quad P_{i,\ell j} = \left[\frac{I_{i,\ell j}K_{i\ell} - K_{ij}}{K_{i\ell} + K_{ij}}\right]_+, \quad I = 1 + I^r + I^{kr} + I^{coop}. \tag{3}$$

For A/B take $\mathcal{H} = \{0, 1, 2\}$ and use zero-density outlet values beyond N. For E retain only r with $i + r \le N$ and renormalize the remaining weights. This finite-domain average is a declared local convention; no imaginary E congestion or extra mass is introduced. Empty donors have zero requests without evaluating incentive ratios. A nonempty donor makes the denominator in (3) positive.

Let $g_i = (N - i)\Delta x$, with $D_r = 0.75$ km. Retain the displayed additive source form, with route and courtesy terms only in the stated directions:

$$\begin{aligned} I^r_{i,EA} &= (1 - g_i/D_r)^3 \quad (0 \le g_i \le D_r), \\ I^{coop}_{i,AB} &= (\rho_{iA} + \rho_{iB})/\rho_{iA} \quad (\rho_{iA} < \rho_c,\ 0 \le g_i < D_r). \end{aligned} \tag{4}$$

These terms are zero otherwise. In free-flow origin cells ($0 < \rho_{i\ell} < \rho_c$), use $I^{kr}_{i,\ell j} = -\rho_{ij}/\rho_{i\ell}$ for a leftward move or a move into rightmost B, and zero otherwise; disable it for the active E route move. In congested origin cells use the source's allowed $I^{kr} = 0$ choice. Equality at ρ_c belongs to that congested branch. The density-mode threshold is nominal and independent of advice. These explicit direction/vacuum/tie and domain conventions are part of the review member, not a recovered author implementation. Route bias is not substituted for a realized merge fraction.

4. Selected allocator, complete update and completion

Mask nonexistent destinations before forming desires. For each nonterminal-E donor, normalize its allowed lateral requests jointly:

$$s_{i\ell} = \sum_{j\neq\ell} P_{i,\ell j}, \qquad b_{i,\ell j} = \frac{P_{i,\ell j}}{\max(1, s_{i\ell})}, \quad b_{i,\ell\ell} = 1 - \sum_{j\neq\ell} b_{i,\ell j}, \quad m_{i,\ell j} = d_{i\ell} b_{i,\ell j}. \tag{5}$$

At last-E set $b_{N,EA} = 1$. All equal reference capacities give equal positive priorities. For each target j, if requests fit its receiving amount, set $z_{i,\ell j} = m_{i,\ell j}$. Otherwise choose the common level λ_{ij} such that

$$z_{i,\ell j} = \min(m_{i,\ell j}, \lambda_{ij}), \qquad \sum_{\ell} z_{i,\ell j} = r_{ij}. \tag{6}$$

This capped sharing returns unused entitlement [5, 6]; its three-lane composition is the selected adaptation. Give unused donor traffic one forward fallback:

$$\begin{aligned} y_{i\ell} &= \min\left\{ d_{i\ell} - \sum_j z_{i,\ell j},\ r_{i\ell} - \sum_h z_{i,h\ell} \right\}, \\ f_{i,\ell j} &= z_{i,\ell j} + \mathbf{1}_{\{j=\ell\}} y_{i\ell}, \qquad n^{k+1}_{i\ell} = n^k_{i\ell} - \sum_j f^k_{i,\ell j} + \sum_h f^k_{i-1,h\ell}. \end{aligned} \tag{7}$$

Set $y_{NE} = 0$, $f_0 = 0$; terminal f_N leaves the cohort and is recorded as completion. No second lateral pass, extra headway multiplier, record history, lossy compression or separate recovery meter is added. Each row sum of f is at most its d, each target sum at most its r. The rule is deterministic but need not maximize unrestricted total flow.

5. An actual same-observation lane-loading witness

Use $v = 60$. In row $N-1$, set all three lane densities to ρ_c; all earlier rows are empty. Last-row densities are

$$X:\ (\rho_E, \rho_A, \rho_B) = (120, 60, 90), \qquad Z:\ (\rho_E, \rho_A, \rho_B) = (120, 90, 60). \tag{8}$$

The entire E profile and A+B profile coincide. Total initial mass is $M_0 = \Delta x(3\rho_c + 120 + 60 + 90) = 153/4$ veh-equivalents in either case. The same observation equality holds for total row/link mass. These are prescribed initial states; no identical past history is asserted.

All last-row donors have $Q(\rho_{N-1,\ell}) = C$ and $D = C$. A/B are both congested: courtesy is off and the declared congested keep-right term is zero. Their route term is zero. The weighted A/B densities are proportional to their current densities because both future outlet values are zero. Thus the only continuing-lane lateral preference in X is $P_{BA} = 1/5$; in Z it is $P_{AB} = 1/5$. E has the selected forced movement to A.

The receiving rates are

$$(R_A, R_B)_X = (2205, 2145), \qquad (R_A, R_B)_Z = (2145, 2205). \tag{9}$$

The selected allocation gives the following *rates*; the update uses these values multiplied by Δt:

	$X :\to A$	$X :\to B$	$Z :\to A$	$Z :\to B$
E	877.5	0	1072.5	0
A	877.5	0	1072.5	450
B	450	1800	0	1755
Total	4005		4350	

For example in X, B's 450-unit request into A is fully served; E and A share the remaining 1755 equally. B has used its whole donor budget, so its unused outlet entitlement does not create more fallback. In Z both receivers fill. E has no forward fallback in either case.

Their next total masses are therefore

$$M_1(X) = \frac{153}{4} - \frac{4005}{600} = \frac{1263}{40}, \qquad M_1(Z) = \frac{153}{4} - \frac{4350}{600} = 31. \tag{10}$$

Hence the reduced observation does not close the actual service or next aggregate evolution. The far-lane redistribution matters; an A-only monotonic intuition is insufficient. This static one-step calculation is not a traffic rollout or an optimal-control result.

For an equal-probability diagnostic pair, let $\mathcal{J}(v, x)$ be the full objective below. Ideal reduced and full-information values are

$$\begin{aligned} V_{red} &= \inf_{v\in[30,60]} \tfrac{1}{2}[\mathcal{J}(v, X) + \mathcal{J}(v, Z)], \\ V_{full} &= \tfrac{1}{2}\inf_v \mathcal{J}(v, X) + \tfrac{1}{2}\inf_v \mathcal{J}(v, Z), \qquad V_{red} - V_{full} \geq 0. \end{aligned} \tag{11}$$

Strict positivity is unproved. One extra scalar ρ_{NA} distinguishes X and Z, so its augmented observer has exactly the full observer's ideal value on this pair. Neither full lane detail nor useful different advice is thereby necessary. Future aggregate flows reveal a distinction after the committed decision; the claim is not absolute unobservability under histories/feedback.

6. Full population-time and eventual unconstrained service

The objective is all remaining population-time from the initialized decision:

$$M_k = \sum_{i,\ell} n_{i\ell}^k, \qquad \mathcal{J}(v, n^0) = \Delta t \sum_{k=0}^{\infty} M_k. \tag{12}$$

Its units are veh-equivalent hours. No later arrivals, uncounted holding stock or residual deletion occurs. Trips complete at f_N, not upon merely arriving at last-E. Action-independent pre-episode travel is outside this remaining-time objective. Traffic after the free exit is outside the member.

Define remaining distance in cell units and a safe residual mass:

$$h_i = N - i + 1, \qquad V(n) = \sum_{i,\ell} h_i n_{i\ell}, \qquad \eta = \Delta x \rho_c = 15/4. \tag{13}$$

Each internal/exit movement advances one row and reduces V by its mass. If Γ_k is the sum of all boundary movement amounts, including exits,

$$V(n^{k+1}) = V(n^k) - \Gamma_k, \qquad M_k \le V(n^k) \le N M_k. \tag{14}$$

Why a nonempty state always makes progress. Take its furthest occupied row. Its receiving row is empty, or consists of exit outlets. Every positive donor has positive d, because $v \ge 30$ and $Q \ge Q_J > 0$. Every allowed receiver there has positive r at zero receiving density. Every donor has a permitted forward/adjacent destination, including E-to-A at the endpoint. Positive equal priorities give a positive realized movement. Thus $\Gamma > 0$ whenever $V > 0$, regardless of the desired fractions.

Why this implies reaching small mass in finitely many steps. Allow the desired fractions b to vary independently over the product of their allowed simplices; this compact set contains every native incentive output. For fixed b, the H budgets and capped-sharing/fallback flow amounts are continuous in (n, b, v), including zero requests. State bounds form a compact set; advice belongs to $[30, 60]$. On its closed subset $V \ge \eta$, Γ is positive everywhere, so it has a strictly positive minimum. Equation (14) therefore forces $V < \eta$ after finitely many updates. This argument does not require the native incentive map itself to be continuous, or an arbitrary convergence tolerance.

Why every movement then receives service. Once $M < \eta$, every physical density and relevant upstream boundary density is below ρ_c. Then $Q = C$ and $R = C$. Total requests into any target are at most

$$\sum_{\ell} d_{i\ell} \le \frac{u_0 \Delta t}{\Delta x} M < C \Delta t = r_{ij}. \tag{15}$$

Thus $z = m$ at every target, no sending cap binds except $v\rho$, and each physical donor advances exactly $c(v) n_{i\ell}$, where

$$c(v) = v \Delta t / \Delta x \in [1/2, 1]. \tag{16}$$

Mass cannot increase, so the regime persists. Consequently $V^{k+1} = V^k - c M_k \le (1 - c/N) V^k$: the remaining mass decays geometrically. Exact finite-time emptying is not required or claimed for all advice values. The proof excludes continued arrivals, blocked exits, zero advice and a changed transport/priority contract.

7. Exact residual value and full regular-branch derivative

Choose any finite index K inside the strict small-mass regime. Since $V_k - V_{k+1} = cM_k$ there, telescope to get

$$\Delta t \sum_{k=K}^{\infty} M_k = \frac{\Delta t}{c(v)} V(n^K) = \frac{\Delta x}{v} V(n^K). \tag{17}$$

This is the exact residual value of the discrete evaluator. A unit of mass in row i has mean remaining time $h_i \Delta x / v$; lane redistribution cannot delay it when all requests receive service. It is not an approximate terminal penalty or a rule deleting small stocks. Combining (12) and (17) gives the finite representation

$$\mathcal{J}(v, n^0) = \Delta t \sum_{k=0}^{K-1} M_k + \frac{\Delta x}{v} V(n^K). \tag{18}$$

Later K inside the same regime gives the same value. For example, with $v = 30$ a small unit of mass in last-E has mean remaining time $\Delta t / c = 0.1/30$ hours, including failed longitudinal advancement due to the advice rate. A unit one row farther upstream takes twice that time.

Let the full state update in (2)–(7) be $F(n, v)$. Suppose a neighborhood of advice preserves a regular finite branch trace through time K. Let $F^{[k]}$ denote its selected smooth arithmetic formula at step k, and write $s^k = \partial n^k / \partial v$. Then

$$s^0 = 0, \qquad s^{k+1} = F_n^{[k]}(n^k, v) s^k + F_v^{[k]}(n^k, v), \tag{19}$$

and the derivative of (18) is

$$\frac{\partial \mathcal{J}}{\partial v} = \Delta t \sum_{k=0}^{K-1} \mathbf{1}^\top s^k + \frac{\Delta x}{v} h^\top s^K - \frac{\Delta x}{v^2} V(n^K). \tag{20}$$

The three terms are finite-trajectory sensitivity, the residual stock's sensitivity, and the direct change in residual travel time. The derivatives $F_n^{[k]}, F_v^{[k]}$ include the source incentives, normalized requests, H budgets, active receiving allocations, fallback and conservation update. They are first derivatives of this complete new update, not the old report's service function or a Hessian. Fixed initialization ties may use the selected branch when all nearby advice follows that same choice; global differentiability in arbitrary state directions is not assumed. Policy parameters can use the subsequent chain rule through advice.

Choose K with strict $M_K < \eta$ and hold that integer fixed for the local derivative. A sufficiently small perturbation retains the exact terminal representation. No derivative of a tolerance-based stopping index is required. The tail already counts all future time and its dependence on v. Differentiating only the finite trajectory would omit the last two terms.

Regularity and events. This result requires a stable finite sequence of native incentive modes and active min/max/allocation constraints. Source courtesy/keep-right changes can still be discontinuous at their state thresholds. Empty donors and structural zero movements have their exact mathematical treatment; no artificial density epsilon is assumed. At genuine switches, two-sided derivatives may fail or finite action changes may cross a jump. The evaluator uses a fixed time grid, so its regular derivative is the discrete chain rule. Continuous-time saltation matrices from the old point model do not apply to this different evaluator. No numerical gradient agreement or useful policy direction has been tested.

8. Scientific limits, comparisons and provenance

The observation ladder uses the same physical rows and forward law: aggregate total; E plus A+B; and all lane states. Add the single adjacent density statistic before attributing a need for all lane detail. The initial pair in (8) establishes service/transition nonclosure, not a positive value in (11). A model-aware reduced controller may reason over the known hidden-state distribution. Histories or repeated actions, if opened, require a fresh information comparison.

For the scalar action, matched-information bounded search and maximum advice are mandatory strong references. A finite search does not certify a global optimum. Separate information value, derivative correctness and decision usefulness. Numerical derivative checks must include the complete residual value; physical validation requires evidence beyond a decrease in this same model's objective. SUMO is optional, as previously directed.

Global assumptions supporting the new proofs. Continuous mass, fixed finite geometry and the common FD; closed initialized cohort and free outlets; once-chosen positive common advice independent of initialization; old-state conservative forward/diagonal movements with positive priorities; completion at the exit and complete population-time; ordinary derivatives only on regular traces. No microscopic order, driver records, integerization, additional recovery law or claim of their necessity is introduced. Continuing arrivals, heterogeneous FD mapping, blocked downstream traffic or different control timing are not covered by these proofs automatically.

Retained and new work. Subraveti [1] is the principal lane-density/incentive framework. The same-author weights [4] clarify its lookahead. Han [2, 3], already part of that congestion lineage, gives the explicit budget/VSL component. Carey/Wright principles [5, 6] support the selected conservative allocation adaptation. The finite-domain conventions, common-FD member, information diagnostic and complete-accounting/derivative arguments are explicitly our review work. The selected composition is not a native published turnkey three-lane model. No source calibration, continuum limit or maximal-throughput theorem transfers.

Status. LF1 and LF2 are user-selected. This bounded qualification is complete for the declared member, delivered for confirmation at JQ1. It does not replace the accepted report automatically, close old pending historical gates, implement a simulator, or claim empirical/gradient success. No new model-family choice is opened by this result.

References

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